

Araz Mehrabani

Mechanical Engineer | Design · CAE · Product Development

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Portfolio: <https://arazmehrabani.github.io/>



WORK EXPERIENCE

Development Engineer 01/2022–02/2023

Paydar | Renewable Energy Systems Tehran

- Designed an electromechanical cleaning system for **PV modules** in **SOLIDWORKS** through manufacturing, assembly, and installation; design optimizations reduced scrap and raw-material use by **approx. 10%**.
- Supported the mechanical design of small wind turbines through component structural assessment and stress analysis in **ANSYS** based on operational load cases.

Mechanical Engineer 03/2017–12/2021

TehranRigzar | Mining and Mineral Processing Equipment Tehran

- Co-developed a high-frequency vibrating screen; replaced the crankshaft drive concept with vibration motors to improve maintainability and reliability.
- Performed FEA-based vibration and resonance analyses in **ANSYS** to evaluate and optimize dynamic behavior; mechanical design in **CATIA V5**.

ADDITIONAL EXPERIENCE

Student Research Assistant — Hakim Sabzevari 12/2014–12/2016

University — Underwater Robotics & Simulation

Junior Project Engineer — Ario Fidar Maham Co. — 11/2012–05/2013

Tehran

EDUCATION

M.Sc. Wind Energy Engineering 03/2023–Present

Flensburg University of Applied Sciences | current grade: 1.9 (German scale)

M.Sc. Mechanical Engineering 09/2014–01/2017

Hakim Sabzevari University | Control Engineering and Robotics

B.Sc. Mechanical Engineering 09/2008–06/2013

Islamic Azad University | Structural Dynamics

SELECTED PROJECT

Autonomous Underwater Welding Robot

Motion control in MATLAB/Simulink, modeling and simulation as part of the M.Sc. thesis; four peer-reviewed publications.

TECHNICAL SKILLS

CAD & Mechanical Design

CATIA V5 · SOLIDWORKS · Siemens NX · Inventor

CAE / Structural Dynamics

ANSYS Workbench · FEA / FEM · vibration and resonance analysis

Product Development

Concept and detailed design · manufacturing drawings · bills of materials (BOMs)

Programming & Data Analysis

Python · MATLAB/Simulink · Power BI

Engineering & Project Tools

MS Office · MS Project · SAP

LANGUAGES

German B1

English C1 / IELTS 7.0

Persian Native

TRAINING

IEEE ANSYS Training

HarvardX Python Programming

SOLIDWORKS Training

CATIA V5 Training

FOCUS AREAS

Product Development & Design

FEA / Structural Dynamics

Design for Manufacturing

Mechanical Engineering & Industrial Equipment